

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 12

(1) Research Program/Project/Study Title: Effect of Different Compost Application Forms on the Growth, Yield and Soil Fertility Dynamics of Kangkong	
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Program or Project has component (if no, proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 2 of 12

(4) Site of implementation	
Municipality/City:	
Province:	
Region:	
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity	(8) Discipline/Field of Specialization
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, significant findings, and implications of the impact of the study.) This study determined the effect of different compost application forms on the growth, yield and soil fertility dynamics of kangkong. The experiment was established in Randomized Complete Block Design (RCBD) and replicated thrice. The different compost forms are Vermicast, F2- Vermi tea, and F3 –Ordinary Compost. Applying vermicast on kangkong plants statistically gave the best growth performance in terms width of widest leaf. Moreover, vermicast also gave the best yield performance in terms of total plot yield, total marketable yield, and total non-marketable yield.	
(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words are already given keywords. This would also include process or methodology employed.) <i>Ipomoea aquatica, rugosa, vermicast, vermi tea, ordinary compost</i>	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Kangkong, also known as water glorybind, water spinach, water convolvulus, and swamp cabbage, is an important green leafy vegetable in Southeast Asia, Taiwan, and	

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 3 of 12

Malaysia. It is found throughout the fresh waters of southern China, and is cultivated in countries such as Ceylon (IFAS Extension, 2021).

Proximate analysis on *Ipomea aquatica* (green kangkong) was carried out using standard methods. The results revealed 51.36% moisture, 42.18% carbohydrate, 1.70% protein, 2.75% ash, 1.20% fiber and 0.81% fat. The investigation of mineral content in micro g/ml showed that the vegetable was relatively high in iron, magnesium and calcium. Other minerals that were present include phosphate, manganese, sulphate and nitrates.

In the country, soil degradation is identified as a significant threat to food security by the National Action Plan (NAP) from 2004 to 2010. It was reported that about 5.2 M hectares are seriously degraded, resulting in a 30 – 50 % reduction in soil productivity and water retention capacity (NAP, 2004). Due to this problem, PCARRD (2006) promotes using organic materials as alternative fertilizer that is affordable and environmentally friendly. One of the powerful and effective technologies widely practiced in organic farming is the use of vermicast or vermicompost.

Vermicompost, an organic fertilizer, is now becoming an exciting approach and proven to be an effective means of improving soil structure, enhancing soil fertility, and increasing crop yield. Plant growth is highly stimulated by vermicompost; because it contains nutrients and other essential nutrients like phosphorus and potassium. Joshi et al. (2014) described vermicompost as an excellent soil amendment and a biocontrol agent through several studies, along with the reasons which make it the best organic fertilizer and more eco-friendly than chemical fertilizers. To evaluate the response of kangkong to the application of various forms of compost, this study was conducted.

The study specifically aimed to:

1. Determine the growth and yield of kangkong as affected by compost applied in various forms;
2. Assess the changes in soil quality indices as affected by compost application in various forms; and
3. Evaluate the nutrient uptake of sweetcorn applied with compost in various forms.

(11) Materials and Methods or Methodology (This includes the study's materials, methods, duration & location. Materials and Methods are headings used in Natural Science research and methodology in Social Sciences. Presentation of materials may not necessarily be in a separate heading but may be incorporated into the specific methods used. Justify paragraphs with main headings in caps.)

The materials and equipment that were used in the study were the following: seeds of sweetcorn, bamboo, fertilizer, insecticide, fungicide, knapsack sprayer, plastic

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 4 of 12

bags, straw lace, bolo, shovel, hammer, nails, digital weighing scale, pen, calculator, tape measure, caliper, data notebook, sacks, and tarpaulin.

Experimental Design. The study was composed of different compost forms: F1 – Vermicast, F2- Vermi tea, and F3 –Ordinary Compost. The study was laid out using the Randomized Complete Block Design with three replications.

Cultural Practices

Land Preparation. Nine plots measuring 4 m x 4 m were prepared to accommodate the 3 treatments replicated. Each block was spaced 1 m while the distance between plots was 0.5 m. Border plots with a width of 1 m were provided. The area was prepared by plowing and harrowing twice with an interval of one week per operation before the construction of plots and paddies.

Securing the seeds. The kangkong seeds were bought from a reputable agricultural supplier in the province.

Sowing of seeds. The kangkong seeds were sown in a seed box containing garden soil, compost, and sand in equal proportion. Seeds were sown in a prepared seed box by evenly spreading them in a finger-made furrow.

Pricking of seedlings. As soon as the seeds germinated or when the radicle protruded, the overcrowded germinated seeds were transferred to the seedling trays to allow them more space to grow and avoid root intertwining. The seedlings were kept in the nursery until they produced three true leaves and a vigorous appearance before being transplanted into the field.

Hardening off seedlings. Seedlings were hardened as soon as they produced true leaves by gradually exposing the seedlings to open sunlight to avoid transplanting shock during transplanting. This was done by exposing the seedlings to open sunlight from early morning until 9:00 a.m. on the first week. The number of exposures to sunlight increased in the following weeks until they produced three or four true leaves and were ready for transplanting.

Transplanting the seedlings. The seedlings with three to four true leaves, vigorous and disease-free, were transplanted into the fields following the recommended planting distance. Transplanting was done late in the afternoon to minimize transplanting shock.

Care of seedlings. The transplanted seedlings were initially covered with banana bracts for four days during extreme sunlight to prevent them from wilting. Plant watering using sprinklers was done every morning and late in the afternoon.

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 5 of 12

Cultivation and weeding. The cultivation of space between rows and hills was done two weeks after transplanting. Shallow cultivation was practiced using a hand trowel to avoid mechanical injury of the roots. Hand weeding was done by pulling the weeds using a trowel as soon as the weeds appeared in the plots.

Watering. The plants were watered whenever necessary, especially when there was no rain.

Application of different forms of compost. Vermicast and ordinary compost were applied to the designated experimental plots in the final stage of land preparation. The application rate was 200 grams per hill based on the recommended rate and analysis. For the vermi tea, the application will be made two weeks before transplanting and every week after that except during the flowering stage. The concentration used was 20 ml/liter of water. The application rate per plot was four liters, and the amount was increased by 1 liter every two weeks.

Pest and disease management and control. Cleanliness and sanitation were strictly implemented in the Project. Pest and disease monitoring were strictly done. In case of high pest and disease incidence, which will threaten the performance of plants, botanical, OHN, and organically formulated pesticides were used. Insect repellent plants were also planted around the experimental areas. The number and kinds of pests that may occur were recorded.

Harvesting/Priming. Kangkong was harvested 30 days after transplanting. Then plants are let to grow for another and harvested after 30 days. This procedure is replicated three times, for a total of 4 harvest.

Data to be Gathered. The following data were gathered: plant height, stem girth, number of leaves, length of the longest leaf, width of the widest leaf, biomass, length of longest root, weight of root, length of the harvested stem, number of marketable plant, number of non-marketable plant, weight of marketable plant, weight of non-marketable part and total plot yield.

(12) Results and Discussions (This part contains the presentation of results (data) concerning the table and figure, to be followed by discussion, with support of reviewed literature. The sequence shall be based on the objectives of the study. Tables are formatted without vertical lines. Horizontal lines are set at the top and bottom portions, below table headings and significant subdivisions. Justify paragraphs with main headings in caps.)

Growth Parameters

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 12

Plant Height

Results revealed (Table 1.0) that tallest kangkong was obtained from F₁ - Vermicast with the mean of 50.06 cm. This was followed by F₃ - Ordinary Compost and F₂ - Vermitea with the means of 49.50 and 47. 50, respectively.

However, the analysis using Analysis of Variance or ANOVA, revealed the plant height of kangkong was not significantly affected by the application of different compost form.

Stem Girth

Result revealed the thickest stem girth of kangkong was obtained from F₁ - Vermicast with the mean of 0.58 cm, followed by F₃ - Ordinary Compost with the mean of 0.54 cm and F₂ - Vermitea with the mean of 0.50 cm.

However, analysis revealed that the stem girth of kangkong was not significantly affected by the application of different compost form.

Number of Leaves

Result revealed that the most number of leaves from kangkong was obtained from F₁ - Vermicast with the mean of 10.30 number of leaves and was followed closely by F₂ - Vermitea with the mean of 10.29 number of leaves. The least among the treatments was F₃ - Ordinary Compost with the mean of 10.0 number of leaves.

However, analysis revealed that the number of leaves of kangkong was not affected by the different compost form.

Length of Longest Leaf

Result revealed that the longest leaf was obtained from F₁ - Vermicast with the mean of 16.23 cm. This was followed by F₃ - Ordinary Compost with the mean of 14.76 cm and F₂ - Vermitea with the mean of 14.01 cm.

Analysis, however, revealed that the length of the longest leaf was not affected by the different compost form.

Width of the Widest Leaf

Result revealed that the widest leaf was obtained from F₁ - Vermicast with the mean of 3.29 cm, followed by F₃ - Ordinary Compost and F₂ - Vermitea with the mean of 3.02 cm and 2.62 cm, respectively.

Moreover, analysis shows that the width of the widest leaf was affected by the different compost form. The highest mean was obtained from F₁ - Vermicast but was comparable to F₃ - Ordinary Compost and the lowest mean was from F₂ - Vermitea.

Biomass

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 12

Result revealed that the heaviest biomass was obtained from F₃ - Ordinary Compost with the mean of 51.07. This was followed by F₁ - Vermicast with the mean of 47.43 cm and F₂ - Vermitea with the mean of 32.27 cm.

However, the analysis shows that the biomass of kangkong was not affected by the different compost form.

Length of Longest Root

Result revealed that the longest root was obtained from F₂ - Vermitea with the mean of 18.29 cm. This was followed by F₁ - Vermicast and F₃ - Ordinary Compost with the mean of 18.03 cm and 17.03 cm, respectively.

However, analysis revealed that the length of the longest root was not significantly affected by the application of different compost form.

Weight of Roots

Result shows that the heaviest roots was obtained from F₃ - Ordinary Compost with the mean of 18.00 cm. This was followed by F₁ - Vermicast with the mean of 16.94 cm and F₃ - Vermitea with the mean of 11.83 cm.

However, the analysis revealed that the weight of the roots was not significantly affected the application of different compost form.

Table 1.0 Mean of different growth parameters applied with different compost form

Different Compost	Plant Height (cm)(ns) (CV=4.50%)	Stem Girth (cm)(ns) (CV=6.39)	Number of leaves (cm)(ns) (CV=4.21%)	Length of the longest leaf (cm)(ns) (CV=11.81%)
F1 - Vermicast	50.06	0.58	10.3	16.23
F2 - Vermitea	47.5	0.5	10.29	14.01
F3 - Ordinary Compost	49.5	0.54	10.0	14.76

Width of the widest leaf (cm)(**) (CV=4.27)	Biomass (g)(ns) (CV=18.15%)	Length of longest root (cm)(ns) (CV=6.05%)	Weight of roots (cm)(ns) (CV=17.69%)
3.29a	47.43	18.03	16.94
2.62b	32.27	18.29	11.83

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 12

3.02a	51.07	17.03	18
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Yield Parameters

Length of Harvested Stem

Result revealed that the tallest harvested stem was obtained from F₁ - Vermicast with a mean of 46.51 cm. This was followed by F₃ - Ordinary Compost and F₂ - Vermitea with a mean of 45.5 cm and 42.2 cm, respectively.

However, the analysis shows that the length of harvested stem was not affected by the application of different compost form.

Number of Marketable Plant

Result revealed that the most number of marketable plant was obtained from F₁ - Vermicast with the mean of 4.86 number of marketable plant. This was followed by F₃ - Ordinary Compost with a mean of 4.56 number of marketable plant and F₂ - Vermitea with a mean of 3.86 number of marketable plant.

However, analysis revealed that the number of marketable plant of kangkong was not affected by the application of different compost form.

Number of Non-marketable Plant

Result revealed that the most number of non-marketable plant was obtained from F₁ - Vermicast with a mean of 1.22 number of non-marketable plant, followed by F₃ - Ordinary Compost and F₂ - Vermitea with the mean of 1.14 and 0.97 number of non-marketable plant, respectively.

However, analysis shows that the most number of non-marketable plant was not affected by the application of different compost form.

Weight of Marketable Part

Result revealed that heaviest marketable part of kangkong was obtained from F₁ - Vermicast with the mean of 45.98 g, followed by F₃ - Ordinary Compost with the mean of 41.36 g and F₂ - Vermitea with a mean of 28.40 g.

Moreover, analysis shows that the weight of marketable part of kangkong was significantly affected by the application of different compost form. The highest mean was obtained from F₁ - Vermicast but was comparable to F₃ - Ordinary Compost, whereas the lowest mean was from F₂ - Vermitea.

Weight of Non-marketable Part

Result shows that the heaviest non-marketable part was obtained from F₁ - Vermicast with a mean of 11.49 g, which was followed by F₃ - Ordinary Compost with a mean of 10.34 g and F₂ - Vermitea 7.10 g.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 12

Moreover, the weight of non-marketable part of kangkong was significantly affected by the application of different compost forms. The highest mean was obtained from F₁ - Vermicast but was comparable to F₃ - Ordinary Compost, and the lowest mean was obtained from the F₂- Vermitea.

Total Plot Yield

Results revealed that the heaviest yield was obtained from F₁ - Vermicast with the mean of 16673 g. This was followed by F₃ - Ordinary Compost with the mean of 14248.67 g and F₂ - Vermitea with a mean of 10146 g.

Moreover, the total plot yield of kangkong was significantly affected by the application of different compost forms. The highest mean was obtained from F₁- Vermicast but was comparable to F₃ - Ordinary Compost, and the lowest mean was obtained from F₂- Vermitea.

Table 2.0 Mean of different yield parameters applied with different compost form

Different Compost	Length of the harvested stem (cm)(ns) (CV=7.36%)	Number of marketable plant(ns) (CV=11.95%)	Number of non-marketable plant(ns) (CV=12.05%)
F1 - Vermicast	46.51	4.86	1.22
F2 - Vermitea	42.2	3.86	0.97
F3 - Ordinary Compost	45.5	4.56	1.14

Weight of marketable part (g)(**) (CV=11.24)	Weight of non-marketable part (g)(**) (CV=11.23%)	Plot yield (g)(**) (CV=11.77%)
45.98a	11.49a	16673a
28.40b	7.10b	10146b
41.36a	10.34a	14248.67a

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 12

conclusions. They may be split or combined based on their length, meaning conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Applying vermicast on kangkong plants statistically gave the the best growth performance in terms of width of the widest leaf, and yield performance in terms of weight of weight of marketable part, weight of non-marketable part and total plot yield. Vermicast or vermicompost is better than ordinary compost due to its higher nitrogen, phosphorus, and potassium content and ability to improve the soil structure and increase water-holding capacity (Singh, 2020).

(14) Output (Indicate the products, processes, or services the Project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes, or services transferred as extension activities)

(17) Acknowledgments (Include only essential persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The researchers would like to acknowledge DA-BAR for the funds.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, hanging indention with 12-point space in between.)

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	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 11 of 12

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	Document Type: DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
		Revision No.	00
	Document Title: COMPLETED RESEARCH	Effective Date	
		Page	Page 12 of 12